**Form 22** Thu. 10 Dec. 2026 Square and cube root; sign of reference functions

**1.** Warming · Flash Calculate $\overline{49}$, $\sqrt{0.25}$ and $(-2)^3$.

**2.** Heart · Method Study the sign of $f(x) = -2x + 6$.

**3.** Heart + · True/false True or false? Justify. For all real $x$, $\sqrt{x^2} = x$.

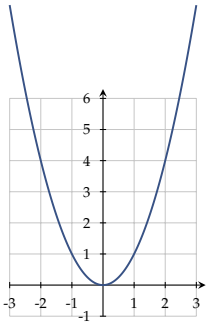
**4.** Back · Flash How many squares compose a cube pattern? A cross (range of 4 squares + 1 square on both sides) is it suitable?

**5.** Red thread · Flash How many faces, peaks and edges have a square-based pyramid?

**Form 23** Mon. 14 Dec. 2026 Graphical and numerical resolution: f $(x) = k$, $f(x) < k$

**1.** Warming · Flash Solve $x^2 = 16$ and $x^2 = -4$.

**2.** Heart · Graph Curve of $f(x) = x^2$ :




Graphically resolve $f(x) = 4$ then $f(x) < 4$.

**3.** Heart + · Method Solve $|x| < 3$ and $|x| = 5$.

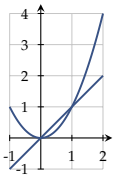
**4.** Back · Trap Calculate $-x^2$ then $(-x)^2$ for $x = -3$.

**5.** Red thread · Method In a triangle $ABC$, $M$ and $N$ are the midpoints of $[AB]$ and $[AC]$. What do we show? What property is quoted?

**Form 24** Mar. 15 Dec. 2026 Resolution of f $(x) = g(x)$ *et* $f(x) < g(x)$

**1.** Warming · Flash Solve 2 $x - 1 = x + 3$.

**2.** Heart · Graph Curves of $f(x) = x$ and $g(x) = x^2$ :




Solve $f(x) = g(x)$ then $f(x) < g(x)$.

**3.** Heart + · Code reading What's sent back `f(-1) == g(-1)` And? `f(0) == g(0)` ?

```
def f(x):
    return x*x
def g(x):
    returns 2*x + 3
```

**4.** Back · Problem Package A: 15,000 Ar fixed + 1,000 Ar per session. Package B: 2,500 Ar per session. To From how many A sessions is it cheaper?

**5.** Red thread · Flash Calculate the midpoint coordinates of the $[AB]$ with $A(-2;5)$ and $B(6;1)$.

## Variable descriptive statistics

**Form 25** game. 17 Dec. 2026 Mean and linearity of mean

**1.** Warming · Flash Average of 12, 15, 9, 14?

**2.** Heart · Method Series 4, 6, 8, 10, 12 averages 8. What is the average if we add 3 to If we multiply each value by 2?

**3.** Heart + · True/false True or false? Justify. The average of two numbers $a < b$ is always strictly between $a$ and $b$.

**4.** Back · Reading In a circular diagram of 80 people, one sector represents 25%. What's his angle?

**5.** Red thread · Flash Calculate PGCD $(84\,;\,126)$ and deduce the irreducible form of $\frac{84}{126}$.

**Form 26** Mon. 4 Jan. 2027 Median and quartiles

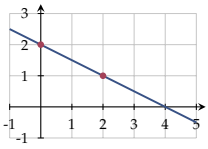
**1.** Warming · Flash Range 7, 2, 9, 4, 5, 11, 3 then give the median.

**2.** Heart · Method Series 3 ; 5 ; 6 ; 8 ; 9 ; 12 ; 14 ; 15. Give $Q_1$, median $Q_3$ and the interquartile gap.

**3.** Heart + · True/false True or false? Justify. At least half of the values in a series are higher or equal to its median.

**4.** Back · Flash Store 0.6, $\frac{5}{8}$ 61% in increasing order.

**5.** Red thread · Graph → formula The right below represents an affine function $f$. Determine $f(x)$.




**Form 27** Mar. 5 Jan. 2027 Standard deviation; effect of adding or deleting a value

**1.** Warming · Flash Calculate $(3-5)^2+(7-5)^2$.

**2.** Heart · Method Series 2 ; 4 ; 6 : mean, variance and standard deviation (value approximated to $10^{-2}$).

**3.** Heart + · True/false True or false? Justify. Add 5 to each value of a series does not change its Standard deviation.

**4.** Back · Construction programme Let $ABC$ a triangle, $I$ the midpoint of $[AB]$ and $D$ the symmetrical of $C$ by relative to $I$. What is the nature of $ACBD$?

**5.** Red thread · Flash $(MN) \parallel (BC)$, $M \in [AB]$, $N \in [AC]$, $AM = 3$, $AB = 5$, $AC = 10$. Calculate $AN$.

**Form 28** 7 Jan 2027 Comparison of series

**1.** Warming · Flash Range of 3, 8, 5, 12?

**2.** Heart · Interpret Two series $A$ and $B$ have even average 12; $\sigma_A = 1.5$ and $\sigma_B = 4$. Which is the most regular?

**3.** Heart + · Interpret $A$ : median 11, interquartile gap 2. $B$ : median 13, interquartile gap 6. Compare the two series.

**4.** Back · Flash We launch two balanced pieces. Likely to get two sides?

**5.** Red thread · Method Program: choose a number; raise squared; add 1. Result for $-3$ ? Can we get 0? Justify.